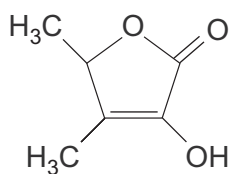


SECTION A*Answer all questions in the spaces provided.*

1. State what is seen, if anything, when aqueous phenol is added to aqueous solutions of the following reagents. [2]

Reagent	Observation
iron(III) chloride	
sodium hydroxide	

2. Sotolone is one of the compounds that is responsible for the smell of raspberries.



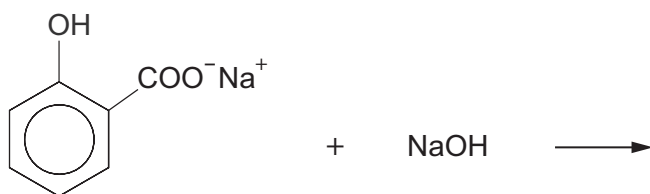
sotolone

Give the **empirical** formula of this compound.

[1]

.....

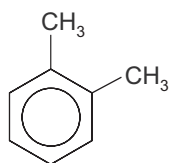
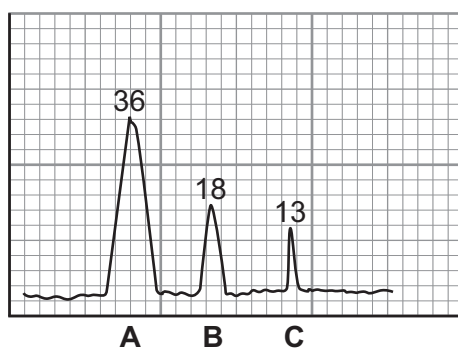
3. Complete the equation for the reaction of sodium 2-hydroxybenzoate with soda lime (NaOH), which, on strong heating, can give phenol as one of the products. [1]



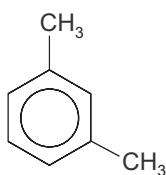
4. State the reagent used to produce phenylmethanol from benzaldehyde. [1]

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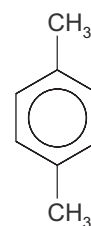
5. The gas chromatogram shows peaks for the products formed by the Friedel-Crafts alkylation of methylbenzene at 0 °C.



A



B



C

Use the data to calculate, to the appropriate number of significant figures, the percentage of 1,3-dimethylbenzene, which is present in the products. [1]

Percentage = %

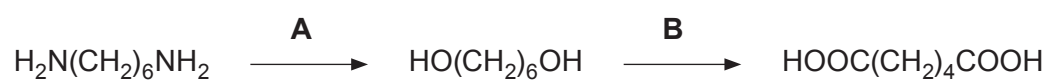
6. Give the systematic name of a compound of formula $C_5H_{10}O_2$ that gives propan-1-ol as one of the products when it is warmed with aqueous sodium hydroxide. [1]

.....



7. Give the displayed formula of the organic compound obtained when cyclohexanol ($\text{C}_6\text{H}_{11}\text{OH}$) reacts with ethanoyl chloride. [1]

8. State the reagents necessary to convert hexane-1,6-diamine to hexane-1,6-dioic acid in two stages. [2]



A

B

